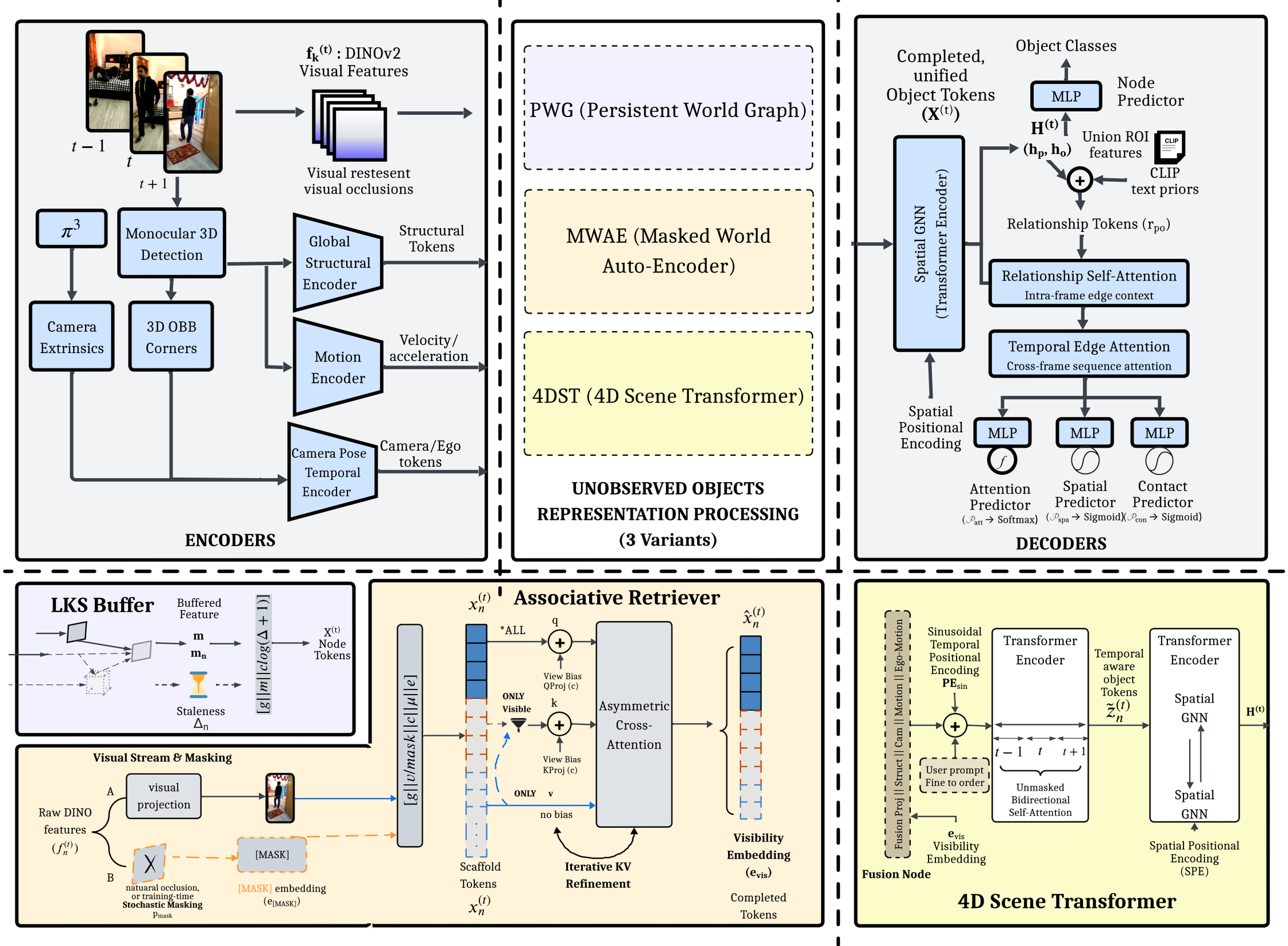
$t-1$
t
$t+1$
$f_k^{(t)}$: DINOv2 Visual Features
Visual restesent visual occlusions
π^3
Monocular 3D Detection
Camera Extrinsics
3D OBB Corners
Global Structural Encoder
Structural Tokens
Motion Encoder
Velocity/ acceleration
Camera Pose Temporal Encoder
Camera/Ego tokens
ENCODERS
PWG (Persistent World Graph)
MWAE (Masked World Auto-Encoder)
4DST (4D Scene Transformer)
UNOBSERVED OBJECTS REPRESENTATION PROCESSING (3 Variants)
Completed, unified Object Tokens ($X^{(t)}$)
Object Classes
MLP
Node Predictor
$H^{(t)}$
(h_p, h_o)
Union ROI features
CLIP
CLIP text priors
Relationship Tokens (r_{po})
Spatial GNN (Transformer Encoder)
Relationship Self-Attention
Intra-frame edge context
Temporal Edge Attention
Cross-frame sequence attention
Spatial Positional Encoding
MLP
MLP
MLP
Attention Predictor ($\mathcal{P}_{att}$ → Softmax)
Spatial Predictor ($\mathcal{P}_{spa}$ → Sigmoid)
Contact Predictor ($\mathcal{P}_{con}$ → Sigmoid)
DECODERS
LKS Buffer
Buffered Feature
m
m_n
Staleness Δ_n
$[g||m||clog(\Delta+1)]$
$X^{(t)}$ Node Tokens
Visual Stream & Masking
Raw DINO features ($f_n^{(t)}$)
A
visual projection
B
natuaral occlusion, or training-time Stochastic Masking
p_{mask}
[MASK]
[MASK] embedding ($e_{[MASK]}$)
$[g||v/mask||c||\mu||e]$
Associative Retriever
$x_n^{(t)}$
*ALL
q
View Bias QProj (c)
ONLY Visible
k
View Bias KProj (c)
ONLY v
no bias
Asymmetric Cross-Attention
Iterative KV Refinement
Scaffold Tokens
$x_n^{(t)}$
$\hat{x}_n^{(t)}$
Visibility Embedding (e_{vis})
Completed Tokens
Fusion Proj || Struct || Cam || Motion || Ego-Motion
Sinusoidal Temporal Positional Encoding PE_{sin}
User prompt Fine to order
e_{vis} Visibility Embedding
Fusion Node
Transformer Encoder
$t-1$ t $t+1$
Unmasked Bidirectional Self-Attention
Temporal aware object Tokens $\tilde{z}_n^{(t)}$
Transformer Encoder
Spatial GNN
Spatial GNN
$H^{(t)}$
Spatial Positional Encoding (SPE)
4D Scene Transformer